

EXP 1B __CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

CODE

```
clc;
clear;
close all;
Am=12;
fm=500;
Ac=5;
fc=5000;
fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
...bplot(3,2,1), plot(t,m), title('Message'), grid on
subplot(3,2,2), plot(t,c), title('Carrier'), grid on
subplot(3,2,3), plot(t,am), title('AM Signal'), grid on
subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on
subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'), grid on
subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}),
title('Power'), grid on
fprintf('μ=%0.2f | USB=%d Hz | LSB=%d Hz | Pc=%0.2f W |
Psb=%0.3f...W\n',mu,USB,LSB,Pc,Psb);
```

OUTPUT

